

Internet Usage & Adoption Analysis Documentation

1. Introduction

This project analyzes internet usage behavior using a structured dataset containing 6,000 records.

The analysis was performed using **SQL, Python (Pandas), and Power BI** to extract reliable insights about adoption trends, demographics, regional differences, and ISP performance.

The objective is to transform raw data into meaningful information that supports **data-driven decisions** in infrastructure planning, pricing strategies, and market expansion.

2. Dataset Overview

The dataset is normalized and contains the following main entities:

- **Users** → demographic information, internet status, daily usage hours, year
- **Regions** → region name, urbanization level, average internet speed
- **ISPs** → internet service providers
- **Tariffs** → monthly cost, speed, connection type

The dataset spans multiple years, enabling time-series growth analysis.

Key Global Metrics

- Total Internet Users: **4,945**
- Internet Penetration Rate: **82.42%**
- Average Download Speed: **26.3 Mbps**
- Average Daily Usage: **3.87 hours**
- Total Monthly Revenue: **\$1,407,490**

3. Methodology

Tools Used

- **SQL:** aggregations, window functions, CTEs
- **Python (Pandas):** validation and exploratory analysis
- **Power BI:** visualization and dashboard insights
- **Mathematical Metrics:**
 - Penetration Rate
 - Growth Rate
 - Moving Average
 - Correlation
 - Ratios & Market Share

All insights produced in SQL were validated using Pandas to ensure correctness.

4. Analysis & Findings

4.1 Internet Adoption Over Time

We measured yearly penetration and growth rate.

Finding

Internet adoption increased steadily across years, but the growth rate fluctuated — indicating a maturing market approaching saturation.

4.2 Demographic Analysis (Age Groups)

- Age group **25–34** has the highest number of users
- Younger groups have higher usage hours
- Older groups show lower adoption but strong potential

Insight

Young users are heavy consumers, while older groups represent future expansion opportunities.

4.3 Usage Behavior

- Average daily usage: **3.87 hours**
- Main purposes:
 - News
 - Streaming
 - Work activities

Mobile internet accounts for **>60%** of connections, confirming the shift toward mobility-first behavior.

4.4 Regional & Urbanization Impact

- **Cairo and Delta** have the highest user concentration
- Urban regions show:
 - Higher speeds
 - Higher daily usage

Insight

Infrastructure quality directly affects adoption and usage intensity.

4.5 ISP Market & Performance

The analysis evaluated:

- Market Share
- ARPU (Average Revenue per User)
- Stability of subscriber base
- Cost vs Speed value

Findings

- Some ISPs maintain high market share with stable growth
- Others show volatility despite large user numbers
- Higher price generally corresponds to higher speed but not always consistently

4.6 Cost vs Speed Relationship

Instead of comparing individual plans, we compared **average ISP cost vs average speed** for statistical validity.

Result

Higher cost tends to provide higher speed, but efficiency varies between providers.

4.7 Growth Potential Identification

Non-internet users were analyzed by region and age group.

Insight

Certain region-age combinations represent strong candidates for future expansion, especially older demographics in less urbanized areas.

4.8 Moving Trends & Stability

- 3-year moving averages were used to smooth adoption trends
- Growth stability analysis identified consistent vs unstable markets

This helps predict future adoption patterns rather than relying on raw yearly fluctuations.

5. Validation

All SQL results were cross-checked using Pandas:

- Aggregations verified
- Growth formulas confirmed
- Correlations validated

This ensured consistency between database querying and analytical modeling.

6. Conclusion

The analysis provides a comprehensive view of internet adoption dynamics:

- Internet usage is high and nearing maturity
- Mobile connectivity dominates behavior
- Urban infrastructure strongly influences performance
- Some ISPs provide better value than others
- Older and rural populations represent major growth opportunities

Combining **SQL accuracy, Pandas validation, and Power BI visualization** produced reliable insights suitable for strategic decision-making in telecommunications planning, pricing, and expansion strategies.